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/ **Practice Exam - 1**

Correct Answer

Next Q

Practice Exam - 1

Question 53 of 117



A 45-year-old female who is status post heart transplant 11 months ago presents to clinic for follow-up. She has recently been battling a upper respiratory tract illness, with runny nose and mild cough but no other complaints. Her immunosuppression consists of tacrolimus (trough 11 ng/ml) and mycophenolate mofetil 1000 mg twice a day. She was weaned off prednisone 2 months ago and has been therapeutic on tacrolimus. She has had donor-derived cell free DNA (dd-cfDNA) monitoring regularly since month 2 after transplant. Her %dd-cfDNA levels have increased in the last three months from 0.08% to 0.15% to 1.8% last week.

The next appropriate step would be to:

- A. Begin valganciclovir and check a serum quantitative CMV DNA with repeat %dd-cfDNA in 1 month for trend
-  B. Obtain an angiogram with IVUS
- C. Begin prednisone burst and taper with repeat %dd-cfDNA in 1 month
-  D. Obtain an expedited endomyocardial biopsy with right heart catheterization

Commentary References

Content Code (Blueprint) - Select the appropriate domain and subdomain from the AHFTC blueprint:

Task (Choose one category: Diagnosis / Management / Test Selection / Interpretation / Prognosis): Test Selection

Teaching Point -: A elevated %dd-cfDNA warrants further investigation for heart graft rejection.

Correct Answer (Best Answer Choice): D

Distractors (3–4 Plausible Incorrect Options):

1. While CMV and other viruses can cause elevations in %dd-cfDNA, the very first priority is ruling out rejection (CMR or AMR).
2. While %dd-cfDNA can be elevated in patients with coronary vasculopathy, the very first priority is ruling out rejection (CMR or AMR). It would be reasonable to evaluate for CAV if the endomyocardial biopsy was unrevealing.
3. There are many things that cause an elevated %dd-cfDNA that are not related to rejection. The %dd-cfDNA is a ratio of donor/recipient circulating cfDNA. From a donor standpoint, the %dd-cfDNA reflects graft injury, which can be due to inflammation (rejection, infection, sarcoid, other), ischemia, toxins, preservation injury, etc. From the recipient perspective, changes in recipient DNA can alter the ratio, leading to increases or decreases in %dd-cfDNA that are independent of graft injury. During rejection, it is felt that both donor (heart injury) and recipient (immune cell increase) cfDNA increase, but donor cfDNA predominates in severe organ injury from rejection. Early after transplant, it is felt that %dd-cfDNA levels are increased due to ischemic reperfusion injury. Additionally, tacrolimus impacts circulating white blood cells (WBC make up >>96% of circulating DNA in humans) and marked variations in tacrolimus levels can impact %dd-cfDNA. As target tacrolimus levels decrease, recipient cfDNA levels can decrease, causing an increase in %dd-cfDNA without change in donor fraction. For this reason, donor quantity (DQ) is often used to complement dd-cfDNA interpretation.

D. Correct answer**Rationale:**

Growing data from various studies including GRAFT, SHORE, Define HT) and others have demonstrated excellent negative predictive value of dd-cfDNA in heart transplant for heart transplant rejection monitoring. Positive results, generally a dd-cfDNA of >0.15-0.20%, warrant evaluation for rejection (AMR or CMR). Studies show that dd-cfDNA elevation can precede histopathological diagnosis of CMR or AMR by 0.5-3.0 months. However, a positive dd-cfDNA does not equate to rejection. There are several factors that can cause a positive result. Thus, biopsy is warranted to confirm rejection and hemodynamic can assist in assessing clinically important histologically silent graft dysfunction (positive dd-cdDNA without a positive biopsy).

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